

IoT and AI-Driven Approaches for Energy Optimization in Off-Grid Solar Systems

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Abstract

The growing reliance on renewable energy sources, particularly solar photovoltaics (PVs), requires intelligent management strategies to address challenges of intermittency, storage, and efficiency in autonomous microgrids. This review investigates IoT-based solutions for energy optimization, focusing on hardware platforms, communication protocols, and intelligent control strategies that enhance the reliability and autonomy of PV-powered systems. This review follows a structured methodological protocol including predefined research questions, database selection, screening criteria, and systematic categorization of studies of IoT-enabled solar microgrid applications, relying on peer-reviewed journal articles, reputable conference proceedings, and scholarly works published between 2020 and 2025. The focus centers on microcontroller-based platforms (e.g., Arduino, ESP32, NodeMCU, TTGO LoRa32) and Single-Board Computers (SBCs) (e.g., Raspberry Pi), alongside the integration of optimization algorithms with Machine Learning (ML) and Neural Network (NN) approaches. Results highlight that lightweight microcontrollers offer cost-effective monitoring, ESP32 and NodeMCU balance real-time analytics with energy efficiency, Raspberry Pi supports edge-level AI processing, and LoRa enables scalable long-range communication for remote PV systems. Furthermore, optimization algorithms (PSO, WOA-SA) and neural models (ANN, LSTM, CNN-LSTM) are explored as methods to improve forecasting accuracy, fault detection, and demand-side management. Conclusions indicate that IoT-based architectures significantly improve energy efficiency, support predictive maintenance, and enable scalable deployment of autonomous solar microgrids. The study emphasizes the necessity of hybrid IoT architectures, combining edge and cloud intelligence, to balance computational complexity, power constraints, and cybersecurity requirements. These findings provide practical insights into designing robust, cost-effective, and scalable IoT-enabled PV microgrids that contribute to decentralized and sustainable energy transitions.



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1. Introduction

The global transition toward sustainable energy systems has accelerated in response to the need to mitigate Greenhouse Gas (GHG) emissions. Conventional power systems are rapidly shifting toward Renewable Energy Sources (RESs), offering cleaner, decentralized,

and more resilient alternatives. Among these, solar photovoltaic (PV) technology has emerged as one of the most prominent solutions due to its scalability, affordability, and environmental benefits [1]. According to the latest Global Status Report, new solar PV capacity additions reached record levels in 2023, confirming solar energy as the fastest-growing renewable technology worldwide [2]. However, the intermittent nature of solar irradiance and the challenges of energy storage and variability still pose significant barriers to the stability and efficiency of PV-based generation [3,4].

The transition from centralized to distributed generation models has led to the emergence of smart grids and autonomous microgrids—localized energy systems that integrate Distributed Energy Resources (DERs) such as PV panels, wind turbines, and battery storage within clearly defined electrical boundaries [5,6]. These systems support bidirectional energy flow, facilitate real-time control, and enable islanded operation in areas with weak or no grid connectivity. Despite their potential, maintaining stability and optimizing energy flow in such distributed environments remain complex tasks due to the fluctuating nature of renewable inputs and varying load demands [3,7].

The integration of the Internet of Things (IoT) into energy infrastructures has become a transformative enabler for achieving intelligent and adaptive management of microgrids. IoT interconnects sensors, actuators, and communication modules to enable real-time data acquisition, monitoring, and control of distributed components [8]. In solar PV-based systems, IoT facilitates predictive maintenance, fault detection, and optimal energy distribution by utilizing data-driven decision-making processes [7,9,10]. Through integration with Artificial Intelligence (AI) and Machine Learning (ML) techniques, IoT frameworks support forecasting of generation and demand, adaptive load balancing, and fault diagnosis, leading to significant improvements in energy efficiency and reliability [3,11,12]. Moreover, cloud and edge computing paradigms enable scalable architectures that combine local intelligence with centralized data analytics, ensuring both responsiveness and system-wide optimization [13,14].

Despite these advancements, IoT-based microgrids face persistent challenges related to security, interoperability, computational limitations, and communication latency. Cyber threats, including False Data Injection (FDI) and Denial-of-Service (DoS) attacks, threaten operational reliability and user trust [5,12,15]. Furthermore, the heterogeneous nature of IoT ecosystems and the resource-constrained nature of embedded devices demand lightweight, efficient, and secure protocols to ensure seamless data flow across distributed environments [8,16].

The objective of this review paper is to provide a comprehensive and structured analysis of IoT-based solutions for energy optimization in autonomous solar microgrids. Specifically, it explores: (a) hardware platforms such as Arduino, ESP32, NodeMCU, and Raspberry Pi; (b) communication protocols and system architectures enabling efficient monitoring and control; and (c) the integration of optimization algorithms and neural network models that support predictive and adaptive energy management. By synthesizing peer-reviewed research from 2020 to 2025, this study highlights the technological advancements, identifies current challenges, and outlines future directions for the development of cost-effective, intelligent, and secure IoT-enabled PV microgrids that support the global shift toward decentralized and sustainable energy systems.

While several recent surveys examine IoT-enabled smart grids or AI-based renewable energy systems, most provide broad conceptual overviews without focusing specifically on implementation-oriented architectures for off-grid solar PV microgrids. In contrast, this review emphasizes hardware-level integration, communication-layer design, and the practical deployment of optimization and neural network models in resource-constrained environments. By structuring the analysis around implementation platforms, algorithmic

strategies, and edge–cloud trade-offs, this work aims to bridge the gap between theoretical surveys and real-world IoT-PV system design. The methodological approach adopted to structure this analysis is described in the following section.

2. Review Methodology

This review follows a structured literature analysis approach focusing on recent technological developments in IoT-enabled off-grid PV systems and autonomous microgrids. The objective is to examine how hardware platforms, communication architectures, and AI-based optimization techniques are integrated to enhance energy efficiency, reliability, and autonomous operation. The majority of the literature was identified through Google Scholar, supported by targeted searches in IEEE Xplore, MDPI, and ScienceDirect. Keyword combinations included terms such as “IoT solar microgrid,” “off-grid photovoltaic monitoring,” “AI-based PV forecasting,” “edge computing in renewable energy,” and “MPPT optimization IoT.” The primary time window considered was 2020–2025, reflecting recent advancements in IoT hardware, LPWAN technologies, and intelligent energy management systems. Selected earlier foundational works were included where necessary for architectural or algorithmic context. Studies were screened based on technical relevance and implementation depth. Priority was given to peer-reviewed journal articles and reputable conference proceedings presenting experimental validation, simulation-based evaluation, or real-world deployments. Conceptual-only studies, purely economic analyses, or works unrelated to off-grid PV applications were excluded. The selected studies were categorized according to hardware platform, communication protocol, optimization strategy, AI model, and deployment architecture (edge, cloud, or hybrid). This structured classification supports the comparative analysis presented in subsequent sections and enables a synthesis beyond purely descriptive reporting.

3. Background and Context

3.1. Renewable Energy Sources and Solar PV

RESs, particularly solar PVs, have become the dominant solution for decarbonizing energy systems and reducing reliance on finite resources [17]. Solar energy is abundant, clean, and increasingly cost-competitive, making it an ideal option for powering remote and off-grid regions. The installation of solar PV systems is driven by policy incentives, technological advancements, and the need to reduce dependence on fossil fuels, providing a feasible and scalable solution for electricity generation [18].

According to the latest global figures, solar power has achieved substantial growth in new capacity additions across the world in 2023. Utility-scale installations are at the forefront of this expansion, followed by commercial and residential systems. China, the United States, and India are among the leading countries driving global growth, while emerging economies are rapidly adopting solar technologies to improve energy access. Moreover, new deployment models such as floating PV systems and agrivoltaics have recently emerged [2].

3.2. Solar PV Systems: Technology, Characteristics, and Integration Challenges

Solar PV systems convert solar irradiance into electrical energy, with performance strongly affected by temperature, shading, and cell characteristics [17,18]. In autonomous microgrids, the inherent variability of PV output makes energy balancing and voltage stability challenging, especially under rapid irradiance fluctuations or partial shading. These conditions require continuous monitoring, efficient storage coordination, and adaptive control strategies to maintain reliable system operation [4,8,11]. Consequently, IoT-enabled sensing and analytics have become essential, providing real-time diagnostics, forecast-

ing, and optimization functions that enhance the performance and resilience of PV-based microgrids [7,9,19].

4. IoT Architecture and Integration in Microgrids

4.1. The Perception Layer: Data Acquisition and Sensing

The Perception Layer forms the interface between the solar microgrid and its physical environment, encompassing sensors, actuators, and signal conditioning circuitry. Its primary function is the reliable collection of critical electrical and environmental parameters essential for real-time Energy Management Systems (EMSs) and Maximum Power Point Tracking (MPPT).

Key Components and Engineering Focus

Electrical Sensing: Accurate measurement of PV voltage and current is fundamental. Integrated current and power sensors (e.g., INA219) are utilized to measure power simultaneously, reducing the need for separate meters [20]. Alternatively, inexpensive Hall-effect sensors (e.g., ACS712) provide proportional voltage data to MCUs [21]. These sensors convert the magnetic field generated by the current into a proportional voltage signal for the MCU.

Cost Optimization: PV performance depends critically on solar irradiance, ambient temperature, and panel temperature. To overcome the high cost of scientific instruments like pyranometers (>\$600), cost-effective solutions substitute them with low-cost illumination sensors (lux meters, e.g., BH1750). The system then implements a data-driven approximation to convert the lux reading to the irradiance value (W/m^2) based on prior calibration [22].

Signal Integrity: Analog signals require conditioning and digitization. Smart sensors (e.g., INA219) often feature a built-in Analog-to-Digital Converter (ADC) and communicate digitally via protocols like I²C (Inter-Integrated Circuit). This layer also integrates actuators (e.g., relays) to convert control signals from the EMS into actions. Minimizing the power consumption of the sensor network is a crucial constraint for maintaining the long-term autonomy of off-grid deployments [20].

4.2. The Network Layer: Communication Stack and Protocol Selection

The Network Layer serves as the critical transport mechanism, ensuring the secure and efficient conveyance of data acquired from the Perception Layer to higher-level systems [11]. The selection of appropriate communication technologies is highly dependent on the energy constraints, data payload size, and extensive range requirements typical of off-grid PV microgrids.

4.2.1. Network Topology and Range Classification

Smart grid communication networks are functionally segmented into hierarchical domains:

- **Local Area Network (LAN):** Handles short-range communication (e.g., 0–100 m) between actuators and the local gateway within a Home Area Network (HAN). Protocols typically include Wi-Fi for higher bandwidth and ZigBee for low-power connectivity [11].
- **Wide Area Network (WAN):** Provides the long-haul link (up to hundreds of kilometers) to a remote cloud or utility server. The primary focus for off-grid PV falls on Low Power Wide Area Network (LPWAN) technologies.

4.2.2. LPWAN Technologies for Wide Area Coverage

LPWAN is indispensable for remote systems, optimizing for long battery life and extended coverage.

- LoRaWAN operates in unlicensed ISM bands and is preferred for its capability to be deployed as a private network at a lower cost. This makes it advantageous for filling coverage gaps in remote areas.
- NB-IoT is a cellular-based 3rd Generation Partnership Project (3GPP) technology in licensed spectrum. It leverages existing mobile infrastructure for reliable, managed connectivity and offers superior-deep indoor penetration [23].

4.2.3. Lightweight Messaging and Security Protocols

The constrained nature of edge devices dictates efficient protocols to minimize overhead:

- Messaging: Message Queuing Telemetry Transport (MQTT) is a publish–subscribe protocol over Transmission Control Protocol (TCP), offering reliability and configurable QoS levels. Constrained Application Protocol (CoAP) is a lighter, RESTful model over User Datagram Protocol (UDP), specifically designed for constrained devices by minimizing overhead and latency.
- Security: Ensuring confidentiality and integrity requires security protocols. Message Queuing Telemetry Transport Secure (MQTTS) utilizes Transport Layer Security (TLS). Conversely, CoAP requires Datagram Transport Layer Security (DTLS) to secure its UDP transmissions, adapting TLS for resource-constrained, unreliable networks.

5. Case Studies and Experimental Implementations

5.1. Microcontroller-Based Implementations and System Architecture

5.1.1. Arduino and Low-Cost Monitoring Systems

Arduino-based architectures remain a cornerstone for educational and small-scale off-grid PV systems due to their simplicity and open-source ecosystem. In [24], Arduino Uno is interfaced with INA219 and DHT11 sensors to measure voltage, current, and temperature. Experimental validation reported average measurement errors below 10%, with voltage errors typically under 1–2% and power errors around 3–4%, depending on operating conditions. Similarly, the authors of [25] implemented a GPRS-enabled Arduino PV monitoring unit for real-time telemetry, offering remote data access and reducing maintenance needs. While limited in computational power, such systems provide low-cost, scalable frameworks for distributed monitoring and logging.

5.1.2. ESP32 and NodeMCU-Based Systems

The ESP32 and NodeMCU microcontrollers bridge the gap between low-power IoT nodes and edge computing. The authors of [22] demonstrated an ESP32-based PV monitoring system with integrated BH1750 irradiance and INA219 power sensors, communicating via MQTT to a cloud server. The system achieved high temporal resolution and energy efficiency. In [21], NodeMCU served as an IoT gateway implementing ML-based regression models and achieved high prediction accuracy close to 99%, supporting informed energy management and planning in small-scale off-grid PV systems.

5.1.3. Raspberry Pi and Edge Intelligence

The Raspberry Pi platform enables real-time edge processing and intelligent control in solar tracking systems. Recent implementations deploy a Raspberry Pi 4B as an embedded edge controller executing AI-based models for autonomous dual-axis solar tracking, achieving up to a 17.5% improvement in annual energy yield compared to static photovoltaic installations while maintaining sub-second control latency [26]. Complementary IoT-based

monitoring architectures utilizing Node-RED, MariaDB, and Grafana support real-time data acquisition, storage, and visualization for system supervision over local networks [27]. The integration of edge AI control with IoT monitoring facilitates autonomous operation and scalable hybrid Edge–Cloud architectures.

5.1.4. LoRa and Long-Range IoT Networks

Long-range communication is essential for remote PV deployments. In [14], the LoRa-based architecture achieved multi-kilometer communication range (up to ≈ 15 km in suburban environments) with high reliability and very low power consumption. As demonstrated in [28], the integration of the LoRa protocol offers a specialized solution for energy-constrained environments. These characteristics enable autonomous operation in rural or off-grid contexts, reducing maintenance frequency and operational costs. LoRa gateways are increasingly paired with ESP32 or STM32 nodes to form private, cost-effective IoT infrastructures.

5.2. Optimization and AI-Enhanced Implementations

5.2.1. Predictive Maintenance and Energy Forecasting

AI-based prediction models integrated into IoT frameworks significantly improve reliability and operational planning. The authors of [29] developed an RNN-based predictive model capable of dynamically adapting control parameters in response to real-time sensor feedback, achieving an 18% reduction in response latency and enabling proactive system maintenance. Similarly, ref. [10] applied CNN–LSTM–Attention models for short-term PV output forecasting, reaching MAPE as low as 1.32%. These models were implemented in cloud environments and later deployed to edge gateways for inference, reducing data transmission overheads.

5.2.2. Digital Twin and Agrivoltaic Applications

A novel Digital Twin approach was presented in [30], where an IoT-connected virtual replica of a PV-agricultural installation simulated system behavior under various shading and tilt configurations. The AI-driven twin achieved real-time feedback optimization with a 22% yield gain and improved agricultural productivity through adaptive energy-agriculture trade-offs. This framework establishes a blueprint for smart hybrid systems combining energy and environmental sensing.

5.2.3. Edge AI for Fault Detection and Efficiency Optimization

As demonstrated in [31], edge devices such as Raspberry Pi and NodeMCU can locally preprocess thermal and visual data from PV modules, enabling lightweight CNN-based classification of dusty, cracked, and partially shaded panels. This edge-level analysis reduces cloud transmission requirements by nearly 50%, while experimental measurements confirm severe efficiency losses—from 15.89 W to 9.07 W under shading and to 5.97 W under dust accumulation. By combining on-site feature extraction with cloud-level analytics, such hybrid IoT architectures enhance real-time reliability and improve the operational efficiency of autonomous PV microgrids.

5.3. Comparative Hardware Benchmarking for Off-Grid PV Systems

Beyond individual case implementations, a structured benchmarking analysis is required to evaluate hardware suitability for autonomous off-grid PV applications. The reviewed studies span microcontroller (MCU)-class platforms (Arduino, ESP32, NodeMCU) and single-board computer (SBC)-class systems (Raspberry Pi 4B), enabling a direct comparison in terms of energy consumption, computational capability, deployment cost orientation, and remote suitability. As summarized in Table 1, this benchmarking framework

synthesizes reported energy consumption profiles, AI inference feasibility, deployment cost orientation, and off-grid suitability into a structured comparative analysis, enabling a clearer evaluation of hardware trade-offs beyond descriptive case reporting.

Table 1. Comparative Benchmarking of IoT Hardware Platforms for Off-Grid PV Systems.

Criterion	MCU-Class (Arduino/ ESP32/NodeMCU)	SBC-Class (Raspberry Pi 4B)	Representative Outcome	Edge AI Capability	Reference
Energy Consumption	Described as low-power monitoring nodes; no multi-watt operational loads reported	Idle: 0.8–2.4 W; Operational avg. ≈4.92 W during AI execution	Sustainable sensing under constrained energy budgets	Limited (cloud-assisted or lightweight models)	[14,22,26]
AI Inference Capability	Primarily rule-based control or regression models; training commonly cloud-based	Full edge deployment of DRL and CNN models; real-time local inference	17.5% annual yield improvement; >95% fault classification accuracy	High (local deep learning execution)	[21,26,31]
Deployment Cost Orientation	Ultra-low-cost hardware emphasis (e.g., minimal component stacks)	Higher hardware provisioning due to memory and processing requirements	Low-cost monitoring vs. AI-enabled optimization	Moderate to High	[22,24,26]
Remote Off-Grid Suitability	Highly suitable due to low power demand and long-range communication (LoRa up to ~15 km)	Suitable with adequate energy provisioning; higher power budget required	Multi-km rural monitoring; autonomous tracking systems	Application-dependent	[14,26]

The benchmarking results reveal distinct trade-offs between MCU- and SBC-class architectures. Microcontroller-based systems prioritize minimal energy consumption and reduced deployment cost, making them particularly suitable for remote and rural installations where energy availability is constrained. However, their computational capacity typically limits on-device intelligence to lightweight regression models or cloud-assisted inference. In contrast, SBC-class platforms, such as the Raspberry Pi 4B, demonstrate significantly higher computational capability, enabling full edge execution of deep learning models, including reinforcement learning and convolutional neural networks. Although these systems exhibit multi-watt operational consumption during AI execution, reported studies demonstrate measurable performance gains such as energy yield improvement and high-accuracy fault detection with low inference latency. Overall, MCU-class platforms are optimal for energy-efficient sensing and telemetry-focused deployments, whereas SBC-class systems are better suited for autonomous control and advanced edge intelligence. Hybrid edge–cloud architectures therefore emerge as a balanced solution, leveraging low-power edge acquisition while reserving computationally intensive training for external servers.

6. Optimization Algorithms and Neural Network-Based Implementations

6.1. Intelligent Optimization in IoT-Based PV Microgrids

The integration of optimization algorithms and neural networks into IoT-based solar PV systems enables dynamic adaptation, fault tolerance, and predictive decision-making. Traditional rule-based control lacks the flexibility required to handle irradiance and load

variability. Thus, hybrid architectures combining IoT sensing, edge computing, and ML facilitate real-time optimization of power flow, storage scheduling, and MPPT [7,10,32].

Optimization strategies in autonomous microgrids pursue multiple objectives: maximizing energy efficiency, minimizing cost, balancing SoC across distributed storage, and maintaining grid stability. Real-time data streams from the perception layer feed optimization modules, where algorithms or trained neural models compute control actions to update inverter setpoints, charge/discharge profiles, and load prioritization policies [33].

6.2. Emerging AI and Optimization Paradigms in IoT-Enabled Solar Energy Systems

Recent developments in IoT-enabled solar energy systems reveal a transition from isolated monitoring solutions toward integrated intelligence-driven architectures. The literature can be categorized into four dominant paradigms: forecasting intelligence, embedded edge AI, control-oriented optimization, and network-level metaheuristic strategies. Forecasting-oriented models primarily employ deep learning architectures such as CNN-LSTM and attention-enhanced recurrent networks to address short-term PV power variability. These approaches leverage long-term historical datasets and centralized training schemes, aiming to reduce forecasting errors and enhance grid stability [10]. Edge AI solutions shift predictive intelligence closer to the sensing layer. Lightweight recurrent neural networks deployed on embedded platforms (e.g., ESP32-based systems) enable real-time inference while reducing communication latency and bandwidth consumption [29]. Such architectures are particularly relevant for autonomous microgrid environments where connectivity may be limited. Control-oriented optimization techniques focus on maximizing energy extraction efficiency through improved MPPT algorithms and converter-level strategies [33]. Although these approaches may not always incorporate deep learning, they remain fundamental for dynamic irradiance adaptation and system stability. Finally, metaheuristic algorithms address energy efficiency from a network perspective. By optimizing cluster-head selection and communication topology in IoT-based WSNs, these techniques aim to prolong network lifetime, reduce load imbalance, and enhance residual energy distribution [34]. Collectively, these paradigms demonstrate that effective IoT-enabled solar systems require multi-layer intelligence integration, combining forecasting accuracy, real-time control efficiency, and network-level optimization.

6.3. Comparative Perspective on AI and Optimization Strategies in IoT-Enabled Energy Systems

A structured comparison of representative AI and optimization methodologies employed in IoT-enabled energy systems is presented in Table 2. The analysis highlights the diversity of modeling approaches, deployment architectures, validation strategies, and performance metrics across recent literature. Deep learning-based forecasting models, such as the CNN-LSTM-Attention architecture in [10], demonstrate strong predictive performance in short-term PV generation forecasting using long-term historical datasets. Embedded intelligence solutions, such as the RNN-based system in [29], illustrate the feasibility of deploying lightweight predictive models on edge devices (e.g., ESP32), enabling real-time monitoring and decision support with reduced latency. Digital Twin implementations [30] extend IoT sensing toward system-level simulation and optimization, enabling scenario-based evaluation of agrivoltaic configurations under controlled environmental conditions. Controller-oriented techniques, such as MPPT optimization algorithms [33], focus on real-time power conversion efficiency under dynamic irradiance conditions. Metaheuristic optimization methods, including the hybrid WOA-SA approach in [34], address energy efficiency at the network level by optimizing cluster-head selection in IoT-based WSNs. Statistical IoT monitoring systems, such as the framework presented in [35], demonstrate the effectiveness of distributed sensing combined with anomaly detection techniques for

photovoltaic fault identification and performance analytics. Similarly, edge-based AI analytics platforms integrating convolutional neural networks and cloud-assisted processing [36] illustrate advanced fault classification capabilities in IoT-enabled PV infrastructures. Overall, the comparative analysis reveals that while deep learning excels in forecasting accuracy, metaheuristic and control-based techniques remain crucial for operational efficiency and system stability. The integration of forecasting intelligence, control optimization, and network-level energy management represents a promising direction for autonomous solar microgrid architectures.

Table 2. Comparative Analysis of AI and Optimization Techniques in IoT-Enabled Energy Systems.

Technique	Deployment Mode	Dataset Scale	Validation Type	Key Metrics	Application Context	Reference
CNN-LSTM-Attention	AI-IoT (centralized DL)	4-year PV data (7.5 min resolution)	Experimental benchmarking	MAPE, MAE, RMSE (MAPE $\approx$ 18.8%)	Short-term PV forecasting	[10]
Embedded RNN	Edge IoT (ESP32 + Cloud)	24 h dataset (15 min interpolation)	Real-time experiment	$R^2 = 0.90$, MAE = 0.48, RMSE = 0.59	Embedded predictive optimization	[29]
Digital Twin + ML	Physical twin + Cloud model	3 experimental campaigns	Controlled laboratory study	22% yield adaptation gain	Agrivoltaic optimization	[30]
MPPT (P&O, IC, Fuzzy-PI)	Converter-level control	Irradiance step scenarios	MATLAB /Simulink simulation	+3–5% efficiency vs. P&O	PV energy conversion control	[33]
Hybrid WOA-SA	Cluster-based IoT WSN	100 m $\times$ 100 m network	MATLAB benchmarking	$\approx$ 25% lifetime gain	IoT network energy optimization	[34]
Statistical IoT Monitoring	Distributed SMD + Server	104 PV panels (18 kW plant)	Field experiment	Outlier detection, power gain	PV fault detection	[35]
AI Remote Sensing Analytics	Edge + Cloud + GPU	Large-scale satellite datasets	Case studies	Accuracy, Precision, Recall	Climate-aware renewable management	[36]

7. Challenges and Open Research Problems

Despite the significant progress in IoT-enabled solar energy optimization, several technical and deployment-related challenges remain insufficiently addressed in the current literature. A critical examination of representative implementations reveals constraints related to computational capacity, dataset adequacy, cloud dependence, scalability, and economic feasibility, particularly in remote and off-grid microgrid environments.

7.1. Computational Constraints and Edge AI Deployment

The feasibility of deploying advanced AI models on low-power embedded hardware remains a central challenge. For instance, the deep reinforcement learning (DRL) framework presented in [26] was implemented on a Raspberry Pi 4B (Raspberry Pi Ltd., Cambridge, UK), featuring a quad-core ARM Cortex-A72 processor and 4 GB RAM, with an average edge power consumption of approximately 4.92 W. Although local inference eliminates cloud latency and enhances autonomy, the study acknowledges hardware wear (e.g., motor degradation) and highlights that large-scale solar farm deployment may require hardware accelerators to sustain performance. Similarly, embedded RNN-based predictive systems [29] claim reduced computational overhead; however, detailed reporting of memory footprint, model size, or energy consumption metrics is often absent. This lack of transparency complicates the evaluation of real-world feasibility in energy-constrained IoT

nodes. These observations indicate that while edge intelligence improves autonomy, model complexity must be balanced against device-level power budgets, thermal constraints, and long-term hardware reliability.

7.2. Dataset Limitations and Model Generalization

Several reviewed studies demonstrate high forecasting accuracy; however, performance is strongly dependent on dataset scale and quality. The CNN–LSTM forecasting model in [10] utilizes four years of historical PV data but reports reduced prediction accuracy for longer forecasting horizons and acknowledges the absence of detailed meteorological inputs. Conversely, the off-grid PV forecasting system in [21] relied on only one month of training data due to temporal constraints, despite recommending a minimum of four months for reliable model training. Such limited datasets increase the risk of overfitting and reduce generalizability under varying climatic conditions. These findings underline a broader issue: many IoT–PV systems are validated under narrow temporal or geographic conditions, raising concerns regarding robustness across seasons, environmental noise, and missing data scenarios.

7.3. Cloud Dependence and Communication Reliability

Although cloud-based architectures facilitate centralized analytics and model retraining, they introduce dependencies that may conflict with off-grid operational objectives. In [21], inference is performed on a server-side MATLAB backend, requiring stable internet connectivity. This dependency may undermine system resilience in rural or remote installations with intermittent network access. Similarly, the LoRa-enabled monitoring framework in [14] relies on cloud services (Google Cloud Platform), which introduces operational costs and long-term service dependence. While LoRaWAN enhances transmission range, extended deployments necessitate sensor calibration and filtering mechanisms to maintain data integrity over time. Hybrid edge–cloud architectures offer a compromise; however, future research must systematically evaluate communication latency, packet loss resilience, and cybersecurity risks in decentralized microgrid contexts.

7.4. Network-Level Energy Optimization and Simulation Gaps

Metaheuristic approaches such as the WOA-SA clustering algorithm in [34] demonstrate significant lifetime improvements in simulated IoT networks. Nevertheless, validation is limited to MATLAB-based simulations over a 100 m × 100 m virtual deployment area with predefined energy parameters (initial energy 0.5 J). Physical implementation constraints, hardware heterogeneity, and real-time communication delays remain untested. This simulation–reality gap suggests that future work should prioritize experimental validation of network-level optimization algorithms in large-scale PV monitoring infrastructures.

7.5. Economic Feasibility and Scalability Considerations

Low-cost monitoring platforms based on ESP32 microcontrollers [22] demonstrate the practicality of affordable IoT sensing solutions. However, detailed lifecycle cost analyses, maintenance overhead evaluations, and long-term reliability assessments are rarely reported. Scalability claims are frequently based on architectural modularity rather than quantitative economic modeling. As PV microgrids expand toward multi-node, multi-sensor deployments, trade-offs between single-board computers (e.g., Raspberry Pi) and microcontroller-based systems must consider not only computational performance but also installation cost, energy overhead, and maintenance complexity.

7.6. Open Research Directions

The analysis above suggests several open research problems:

- Development of energy-aware AI model compression techniques tailored for edge IoT-PV nodes.
- Standardized benchmarking frameworks combining dataset size, hardware specifications, energy consumption, and validation protocols.
- Experimental evaluation of hybrid edge–cloud architectures under real off-grid connectivity conditions.
- Long-term economic feasibility studies integrating CAPEX, OPEX, and lifecycle degradation factors.
- Robustness testing under seasonal variability, sensor drift, and incomplete datasets.

Addressing these challenges is essential for transitioning IoT-enabled solar microgrids from experimental prototypes toward resilient, scalable, and economically sustainable deployments.

8. Conclusions

This review presented a comprehensive synthesis of IoT-based approaches for energy optimization in autonomous solar microgrids, emphasizing the integration of sensing, communication, and intelligent control. The convergence of IoT, artificial intelligence, and optimization algorithms provides the technological foundation for transforming conventional PV installations into self-learning, adaptive energy systems.

At the architectural level, IoT frameworks were shown to enable real-time data acquisition, system interoperability, and scalability through layered designs that combine low-cost microcontrollers, LPWAN communication, and cloud-edge collaboration. Optimization algorithms such as PSO, GA, and Fuzzy Logic improve energy efficiency and demand-side management, while neural models including ANN, LSTM, and CNN–LSTM enhance forecasting accuracy and predictive maintenance capabilities. Experimental implementations confirmed that combining edge analytics with cloud-based intelligence achieves measurable improvements in performance, efficiency, and resilience.

Despite these advancements, challenges persist in cybersecurity, interoperability, and energy constraints. Future research should focus on lightweight AI deployment, federated learning, and green IoT design to reduce the computational and environmental footprint of large-scale microgrid systems. The integration of digital twins and AI-driven cybersecurity frameworks will be crucial for ensuring operational reliability and long-term sustainability.

In conclusion, IoT-based intelligent optimization constitutes a pivotal enabler for the global energy transition. By merging data-driven intelligence with renewable technologies, autonomous solar microgrids can deliver decentralized, efficient, and sustainable energy solutions, supporting the broader vision of resilient smart energy infrastructures aligned with net-zero objectives.

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Abbreviations

The following abbreviations are used in this manuscript:

3GPP	3rd Generation Partnership Project
ADC	Analog-to-Digital Converter
AI	Artificial Intelligence
ANN	Artificial Neural Network
ARM	Advanced RISC Machines
CAPEX	Capital Expenditure
CNN	Convolutional Neural Network
CNN-LSTM	Convolutional Neural Network-Long Short-Term Memory
CoAP	Constrained Application Protocol
DERs	Distributed Energy Resources
DoS	Denial-of-Service
DRL	Deep Reinforcement Learning
DTLS	Datagram Transport Layer Security
EMS	Energy Management System
FDI	False Data Injection
GA	Genetic Algorithm
GHG	Greenhouse Gas
GPRS	General Packet Radio Service
GPU	Graphics Processing Unit
HAN	Home Area Network
I2C	Inter-Integrated Circuit
IoT	Internet of Things
ISM	Industrial, Scientific, and Medical
LAN	Local Area Network
LoRa	Long Range
LoRaWAN	Long Range Wide Area Network
LPWAN	Low Power Wide Area Network
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MCU	Microcontroller Unit
ML	Machine Learning
MPPT	Maximum Power Point Tracking
MQTT	Message Queuing Telemetry Transport
MQTTS	Message Queuing Telemetry Transport Secure
NB-IoT	Narrowband - Internet of Things
NN	Neural Network
OPEX	Operational Expenditure
P&O	Perturb and Observe
PSO	Particle Swarm Optimization
PV	Photovoltaic
QoS	Quality of Service
RES	Renewable Energy Sources
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network

SBC	Single-Board Computer
SMD	Surface-Mount Device
TCP	Transmission Control Protocol
TLS	Transport Layer Security
UDP	User Datagram Protocol
WAN	Wide Area Network
WOA-SA	Whale Optimization Algorithm Simulated Annealing
WSN	Wireless Sensor Network

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